Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours Max. Marks: 35

Q1. Describe the steps to download and install VirtualBox on a Windows machine. Include the link for downloading and any prerequisites that need to be checked before installation. (15)

Q2 While analyzing a malware sample, you notice that it spawns multiple processes and performs unusual network connections. How would you use Process Hacker or Process Monitor to track and analyze these behaviors in real-time? (15)

Q3 Viva (05)

Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours Max. Marks: 35

Q1. Describe the steps to download and install VM Ware on a Windows machine. Include the link for downloading and any prerequisites that need to be checked before installation. (15)

Q2. After performing dynamic analysis of a sample using Noriben, Process Monitor, and Process Hacker, how would you document and report the findings? What key elements should be included in your analysis report? (15)

Q3. Viva (05)

Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours Max. Marks: 35

Q1. Describe the steps to install IDA Pro on a Windows machine. Include any prerequisites or configurations needed during the installation process. (15)

Q2. After installing VMware/VirtualBox, what are the key configuration settings you should verify before creating a new virtual machine? (15)

Q3. Viva (05)

Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours Max. Marks: 35

Q1. How would you create a new Linux VM in VirtualBox, and what steps would you follow to allocate resources like RAM, CPU, and disk space for optimal performance? (15)

Q2. How would you open a malicious DLL file in IDA Pro? Describe the initial analysis steps to identify the main function of the DLL and any potentially dangerous operations. (15)

Q3. Viva (05)

Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours Max. Marks: 35

Q1. What steps would you follow to create a Windows 10 VM in VMware Workstation Player? Explain how to configure system resources (such as CPU cores, RAM, and disk space) for this VM. (15)

Q2. Write a simple IDAPython script that extracts the list of all functions in a given malware binary. How would you modify the script to list functions that are suspicious or related to common malware behavior (e.g., system calls)? (15)

Q3. Viva. (05)

Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours Max. Marks: 35

Q1. How do you install the xxd, file, md5sum, sha256sum, and sha1sum tools on a Linux-based system? Provide the commands for each (15)

Q2. Describe the process of downloading a sample malware file for static analysis. What precautions should you take when downloading and handling malware files? (15)

Q3. Viva (05)

Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours Max. Marks: 35

Q1. How do you track down malicious DNS requests using Wireshark, and what abnormal patterns would you look for? (15)

Q2. In IDA Pro, you find a suspicious function that could be related to network communication or file modification. How would you analyze this function in detail using both IDA Pro's interactive features and IDAPython scripts to understand its role in the malware's behavior? (15)

Q3. Viva (05)

Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours Max. Marks: 35

Q1. How would you use the md5sum, sha256sum, and sha1sum commands to generate cryptographic hash values for a malware sample? Explain the importance of hash values in malware analysis. (15)

Q2 How do you install x64dbg on a Windows system? What dependencies, if any, need to be installed beforehand, and what configurations should be considered during installation? (15)

Q3. Viva (05)

Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours   Max. Marks: 35

Q1. After generating the cryptographic hash values for the malware file, how would you verify if the malware sample has been previously identified by any antivirus databases? (15)

Q2. After opening a malicious binary in x64dbg, describe the initial steps for setting up the debugger. How do you configure the debugger to avoid common traps like anti-debugging techniques? (15)

Q3. Viva (05)

Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours | Max. Marks: 35

Q1.How would you load a malicious DLL into x64dbg for debugging? What are the essential steps to ensure that the debugger is attached properly to the DLL? (15)

Q2. How would you scan a suspicious binary file using VirusTotal? Walk through the steps, including uploading the file and interpreting the results (15)

Q3. Viva (05)

Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours Max. Marks: 35

Q1. Explain how to use Metadefender to scan and analyze a suspicious binary file. How does Metadefender's analysis compare to other tools like VirusTotal? (15)

Q2. Explain the process of opening and analyzing a sample malware executable in x64dbg. How do you identify key indicators like anti-debugging techniques, unpacking routines, or API calls to analyze further? (15)

Q3. Viva (05)

Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours   Max. Marks: 35

Q1. How would you use the strings command to extract human-readable strings from a binary file? Provide an example command and explain what kind of information you might extract from the binary.

(15)

Q2. How do you load and analyze a malicious DLL in IDA Pro? What steps do you take to identify the main entry points and key functions within the DLL? (15)

Q3. Viva (05)

Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours Max. Marks: 35

Q1. Write a Python script that accepts a suspicious malware file, prints its file type using the file tool, and outputs the cryptographic hash values (MD5, SHA1, SHA256) for the file. (15)

Q2. Describe the most common entry points for malware on a computer system. Provide examples of how malware can exploit these entry points (e.g., email attachments, software vulnerabilities, social engineering). (15)

Q3. Viva (05)

Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours Max. Marks: 35

Q1. After scanning a suspicious file using VirusTotal, how would you interpret the results and decide whether the file is malicious or benign? What steps would you take next if the file is flagged as suspicious or malicious? (15)

Q2. How can you use Wireshark to analyze network traffic and detect phishing attempts or access to fake websites? (15)

Q3. Viva (05)

Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours Max. Marks: 35

Q1. How do you configure a virtual machine (VM) for dynamic malware analysis? What are the key software and tools you need to install for this type of analysis (15)

Q2. How would you secure a network against malware that uses phishing or malicious links to enter through email? What layers of defense (such as email filtering, firewalls, and endpoint protection) should be implemented? (15)

Q3. Viva (05)

Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours Max. Marks: 35

Q1. Why is it important to create snapshots of your VM before performing dynamic analysis on malware, and how can you restore the VM if something goes wrong? (15)

Q2. How can you identify unusual network traffic patterns in Wireshark that may indicate a network is infected? (15)

Q3. Viva (05)

Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours　　　　Max. Marks: 35

Q1. What precautions should you take when setting up a lab environment for malware analysis? Discuss key configurations like isolation, snapshots, and network settings. (15)

Q2. Walk through the steps to detect a spam email using various email client features or third-party tools. How would you identify signs of a phishing email in a typical spam message? (15)

Q3. Viva (05)

Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours Max. Marks: 35

Q1. How do you install and configure Process Hacker on your analysis machine? What are the essential features of Process Hacker that are useful for analyzing malware behavior? (15)

Q2. Explain how email filtering works to detect and block spam emails. What are some common techniques used by email filters to categorize emails as spam or legitimate? (15)

Q3. Viva (05)

Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours Max. Marks: 35

Q1. After installing Process Hacker, describe the process of using it to analyze a sample malware. What specific indicators in the Process Hacker interface should you focus on (e.g., processes, threads, modules)? (15)

Q2. Describe the common techniques used in phishing attacks. What are the key indicators that an email or website might be part of a phishing attack? (15)

Q3. Viva (05)

Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours Max. Marks: 35

Q1. What steps would you take using Process Hacker to monitor system activities like process creation, file system changes, and network connections made by a suspicious malware sample? (15)

Q2. How can you use a browser extension or security tool to protect yourself from phishing websites? Explain how these tools can help identify fraudulent sites. (15)

Q3. Viva (05)

Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours Max. Marks: 35

Q1. How do you install Process Monitor, and what configurations should you adjust to effectively capture the activities of malware during dynamic analysis (15)

Q2. How would you respond if you received a phishing email that appeared to be from your bank or employer? What steps should you take to report and avoid being compromised? (15)

Q3. Viva (05)

Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours Max. Marks: 35

Q1. After installing Process Monitor, explain how to filter out unnecessary noise and focus on relevant system activity while analyzing a sample malware. What types of events should you be looking for? (15)

Q2. How would you configure a Windows machine to block unauthorized USB devices while still allowing approved devices to function? What steps would you take to set up a device management policy for USB ports? (15)

Q3. Viva (05)

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T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours Max. Marks: 35

Q1. How would you use Process Monitor to identify system file or registry modifications that a malware sample may make? Give an example of an event you would investigate in detail. (15)

Q2. In a situation where a USB drive is suspected to contain malware, how would you safely analyze the device on an isolated system without risking contamination of the main network? What tools or techniques would you use to check for malicious files? (15)

Q3. Viva (05)

Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours Max. Marks: 35

Q1. What is the Noriben tool, and how is it different from other dynamic analysis tools like Process Monitor or Process Hacker? Describe how you would install and set up Noriben for use (15)

Q2. How do you set up REMnux on a virtual machine? What are the key installation steps, and how would you ensure that the system is ready for malware analysis after installation? (15)

Q3. Viva (05)

Savitribai Phule Pune University

T.Y. B.Sc. (Cyber and Digital Science) Sem-VI

(CDS-368 A Lab on CDS-367A-Malware Analysis)

Time: 3 Hours | Max. Marks: 35

Q1. Explain the role of rundll32.exe in Windows and how it can be used to execute a DLL file. Why might malware use rundll32.exe for execution? (15)

Q2 How would you use REMnux to perform a basic static analysis of a sample malware file? Which tools within REMnux would be used for tasks like file type identification, hashing, and disassembling? (15)

Q3. Viva (05)